

ALBERT JOSHWA A

Java Backend Engineer | Microservices & Distributed Systems | AI / LLM Systems

+91 99442 70690 | albertjoshrock101@gmail.com | Coimbatore, Tamil Nadu, India

linkedin.com/in/albert-joshwa-625511208 | github.com/AlbertJoshwa1802011

PROFESSIONAL SUMMARY

Java backend developer with 4+ years building enterprise SaaS and FinTech products at Zoho Corporation and GoFrugal Technologies (Zoho group). Delivered 15+ API decompositions in Zoho Books' monolith-to-microservices program, now serving 6+ Zoho Finance products, and built a real-time bidirectional sync engine live with paying customers on the ONDC Network after clearing internal security and DPIA review. Java 17, Kafka, REST and event-driven design, delivered in Agile/Scrum sprints with cross-functional teams.

TECHNICAL SKILLS

Languages: Java (11, 17, 21), Python, JavaScript, SQL, Deluge

Backend Frameworks: Spring Boot, Spring Cloud, Spring Security, Spring Data JPA, Hibernate, Spring MVC, JDBC, Undertow

Architecture: Microservices, Monolith Decomposition, REST API Design, Event-Driven Architecture, Webhook-Driven Services, RMI, Multi-Tenant SaaS

Data & Messaging: MySQL, PostgreSQL, Cloudflare D1, Redis, Apache Kafka, Schema Migrations, Query Optimization

AI & Agentic Systems: Large Language Models (LLM), Multi-Agent Orchestration, Prompt Engineering, Model Context Protocol (MCP), Human-in-the-Loop AI

Testing, Security & Quality: JUnit 5, Mockito, Test-Driven Development (TDD), Integration Testing, SAST Scanning, Secure Code Review, OAuth 2.0, HMAC Webhook Verification, Role-Based Access Control

Cloud & DevOps: AWS (EC2, S3), Docker, Kubernetes, Cloudflare Workers and Pages, Gradle, Maven, Git, GitLab, CI/CD Pipelines

Domain: FinTech, ERP, Accounting and Billing SaaS, Point of Sale, Multi-Tenant Platforms

PROFESSIONAL EXPERIENCE

Member Technical Staff - Java Backend Engineer

Jul 2023 - Present

Zoho Corporation (SaaS suite used in 150+ countries) - Customization Platform, Zoho Finance | Chennai / Coimbatore, India

- Owned the decomposition of 15+ tightly coupled Zoho Books APIs (Java 17) into independently deployable services as part of the Finance platform's monolith-to-microservices program, now supporting 6+ product lines including Zoho Books, Inventory, Expense, Payroll and Billing.
- Engineered a connector-based plug-and-play architecture that resolves cross-repository dependencies at startup, enabling each of the 6+ Finance products to build, deploy and release independently without coordinated downtime windows; migrated 8+ shared entity classes into a common module through a phased rollout across product build cycles to avoid regressions.
- Solved cross-service data dependencies with two complementary strategies: static metadata injected through HTTP headers at request initiation, and RMI for dynamically generated data inside the microservice layer.
- Built a real-time bidirectional sync engine between GoFrugal On-Premise (Vikra ONDC) and Zoho Finance applications using incremental sync and blob-based data processing; cleared internal security review and DPIA audit and is now in use by paying customers on the ONDC Network.
- Architected that sync framework around a generic connector pattern, letting the follow-on Zoho Commerce integration reuse the same extensible design and complete in 50% less time than the initial GoFrugal integration.
- Designed and shipped the System Module Layout Customization feature for the Vendor and Sales Receipt modules, letting end users reorder and restructure field layouts across 6+ Finance products.
- Independently designed and built the Zoho Finance Developer Console in under one month, a JavaScript internal tool that unifies Gradle builds, server startup and sub-repository management into a visual workflow builder; adopted by 15+ engineers on the team.
- Identified and remediated 10+ security vulnerabilities, including a permission-level escalation in document upload workflows, through SAST scanning and manual code review ahead of each production deployment.
- Authored unit and integration tests (JUnit 5, Mockito) for migrated microservice modules, improving regression coverage across the shared platform layer.

Member Technical Staff

Jun 2022 - Jul 2023

GoFrugal Technologies (Zoho group) - Connect Team, Alerts | Chennai, Tamil Nadu, India | Intern to full-time conversion

- Integrated the Meta WhatsApp Business API into GoFrugal On-Premise and On-Cloud POS systems; the integration now ships as part of the GoFrugal product used by 1,000+ merchants across their network.
- Launched the Alert Gateway microservice on Undertow and Java 17 with Apache Kafka as the message broker, delivering asynchronous WhatsApp and SMS notifications from a single consolidated service deployed on AWS EC2 and S3.
- Built a template registration UI with rich text formatting and live preview, letting business users create and manage WhatsApp notification templates without engineering involvement.
- Remediated 5+ critical vulnerabilities (SQL injection, path traversal, XSS) through security analysis within Agile sprints, hardening every module exposing open API endpoints.

PROJECTS

Church Contribution Portal - Full-Stack Payments Platform

Dec 2024 - Present

Live in production | light-of-jesus-ministry-contributions.pages.dev | Cloudflare Pages Functions, D1, Razorpay

- Designed and shipped a live contribution management platform handling real payments: 23 REST endpoints running as Cloudflare Pages Functions over a Cloudflare D1 database of 20 tables, evolved through 13 additive, backward-compatible migrations with zero production data loss.
- Automated the money path end to end with a Razorpay checkout plus a webhook receiver that verifies every callback with HMAC-SHA256 signature validation before recording a contribution, replacing manual spreadsheet reconciliation.
- Enforced role-based access control across 12 granular permissions and shipped a flag-gated v2 rollout behind a signed-cookie beta allowlist, so new flows reach opted-in users while production stays on the stable path.
- Hardened the platform with a 300+ case automated test suite that exercises the API handlers against a real in-memory SQLite instance, with CI blocking deployment on a red suite; migrated the original Google Apps Script backend onto Cloudflare with no downtime.

Platform Copilot - AI Multi-Agent Workflow Automation

May 2026

Zoho Books Internal Hackathon

- Architected a two-agent LLM pipeline (Pattern Detector plus Executor) that reads Zoho Books activity logs, detects repetitive business patterns across 5 categories and auto-generates production-ready workflow automations with human-in-the-loop approval.
- Designed an enriched prefill JSON schema with a topological dependency map, enabling cross-agent coordination across 6 automation primitives: email templates, alerts, field updates, custom functions, webhooks and push notifications.
- Delivered a working production baseline, an active Zoho Books workflow with 3 wired actions (custom function, email alert, webhook) verified in a live org, and authored a hybrid MCP and UI playbook that cut manual workflow builds from 60 minutes to 5.
- Iterated the Executor system prompt across 3 versions against live API behaviour, diagnosing and documenting an MCP wrapper defect that silently dropped inline action objects (error 107027), then rewriting the agent around a pre-create-then-reference pattern.

TradeSync POS - Multi-Platform Business Management Suite

Oct 2024

github.com/AlbertJoshwa1802011/allwin-traders-pos-main

- Developed a 3-platform business management suite comprising a voice-assisted Android delivery app, a Vercel-hosted web invoice manager and a customer-facing storefront, all synchronising in real time through Google Firebase on a multi-tenant database architecture.

EDUCATION

Bachelor of Engineering, Electronics and Communication Engineering

2018 - 2022

Sri Ramakrishna Engineering College, Coimbatore, Tamil Nadu | CGPA 8.1 / 10